

Evaluating Neural Weather Forecasting Models During Extreme Weather Events

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INTRODUCTION PROBLEM

It is crucial to develop timely and accurate weather forecasts.

Global **climate change** has led to increases in the **frequency** and **severity** of **extreme weather events**

As these events are exacerbated by climate change beyond our natural climate variability, they become even harder to predict.

ML MODELS

Struggle to generalize to extreme weather events due to their **limited representation** in available data.

AI methods result in **smoother forecasts**, underestimating the magnitude of extreme weather events

Extreme events are driven by **complex interactions** not in simplified and controlled general weather benchmarks.

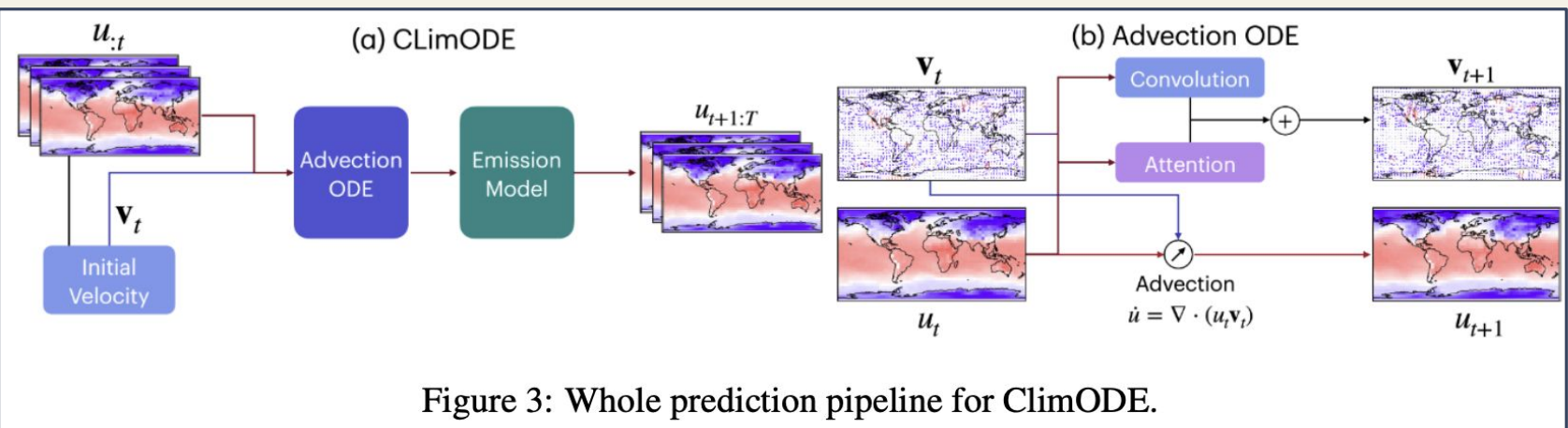
Uncertain whether models can reliably forecast weather patterns amidst the unpredictable climate changes anticipated.

OBJECTIVES

How can the performance of ML models on extreme weather events be improved?

- How do deterministic SOTA ML models generalize to extreme weather events?
- Can fine-tuning with extreme weather data improve generalization?

RESEARCH MODEL: ClimODE

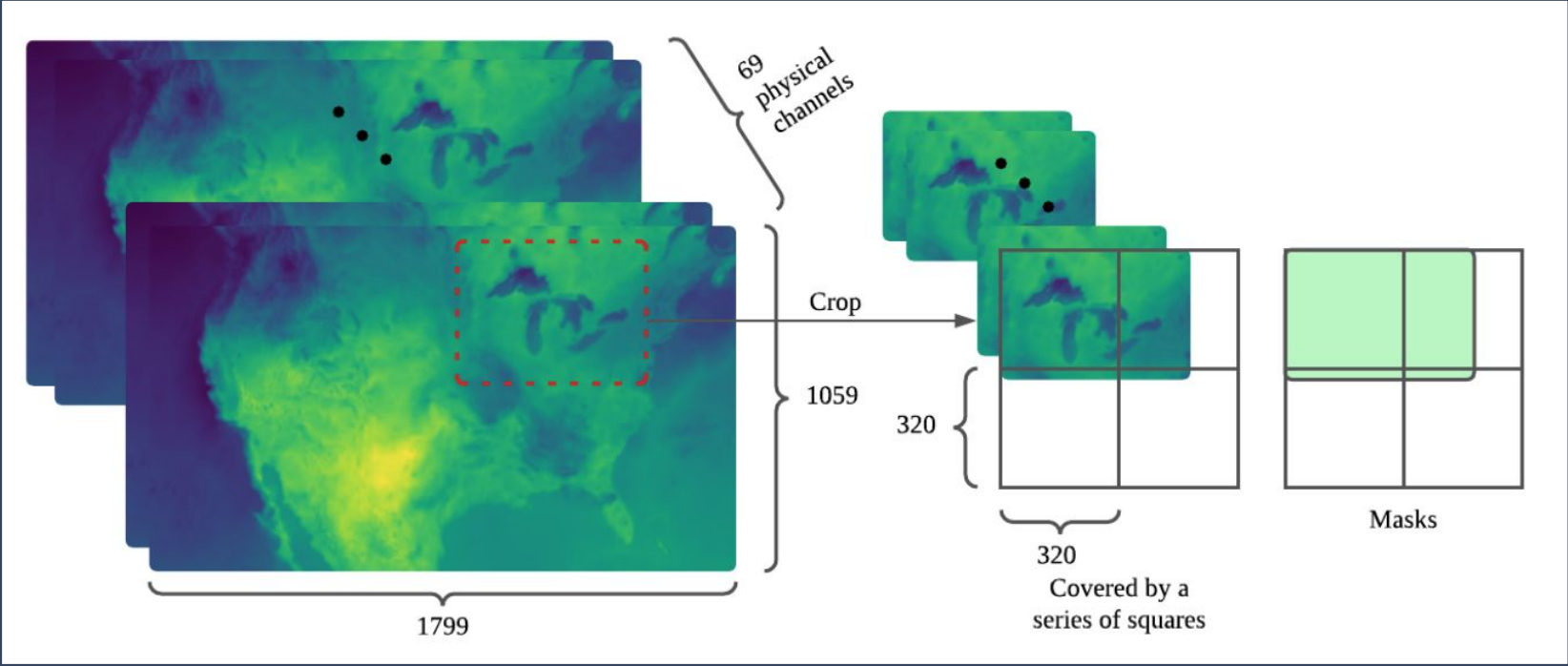


A spatiotemporal continuous-time PDE climate and weather model.

Models precise weather evolution with **physics-conserving** dynamics, which could increase model **consistency with the physical world**, especially on unseen or extreme events.

More **computationally efficient** than other SOTA models and can be trained on a single GPU

DATASETS



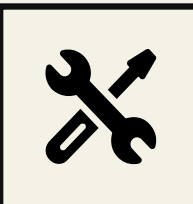
HRRR: High-Resolution Rapid Refresh (Training)

Hourly high-resolution US dataset
Data extraction: selected random **320x320** crops from **1799x1059** grid of the US across random timeframes.

HR-Extreme (Evaluation & Fine-tuning)

New weather dataset specifically focused on 17 extreme weather types
3-hour window of data: 2-hour prior to event and event's hour

METHOD



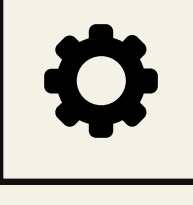
Adapting ClimODE to high-resolution weather data



Pre-training on standard weather (HRRR)



Evaluating on extreme events (HR-Extreme)



Fine-tuning on extreme-events

ADAPTING ClimODE

Batched Processing: Adapted ClimODE to process HR-Extreme's independent 3-hour windows

Local Smoothing: Replaced kernel-based smoothing with local smoothing for scaling to high-resolution.

$$\hat{v}_k(t) = \arg \min_{v_k(t)} \left\| \tilde{u}_k(t) + v_k(t) \cdot \tilde{\nabla} u_k(t) + u_k(t) \tilde{\nabla} \cdot v_k(x, t) \right\|_2^2 + \alpha \left\{ \|v_k(t)\|_K^2 + \|\nabla v_k(t)\|_2^2 \right\}$$

METRICS

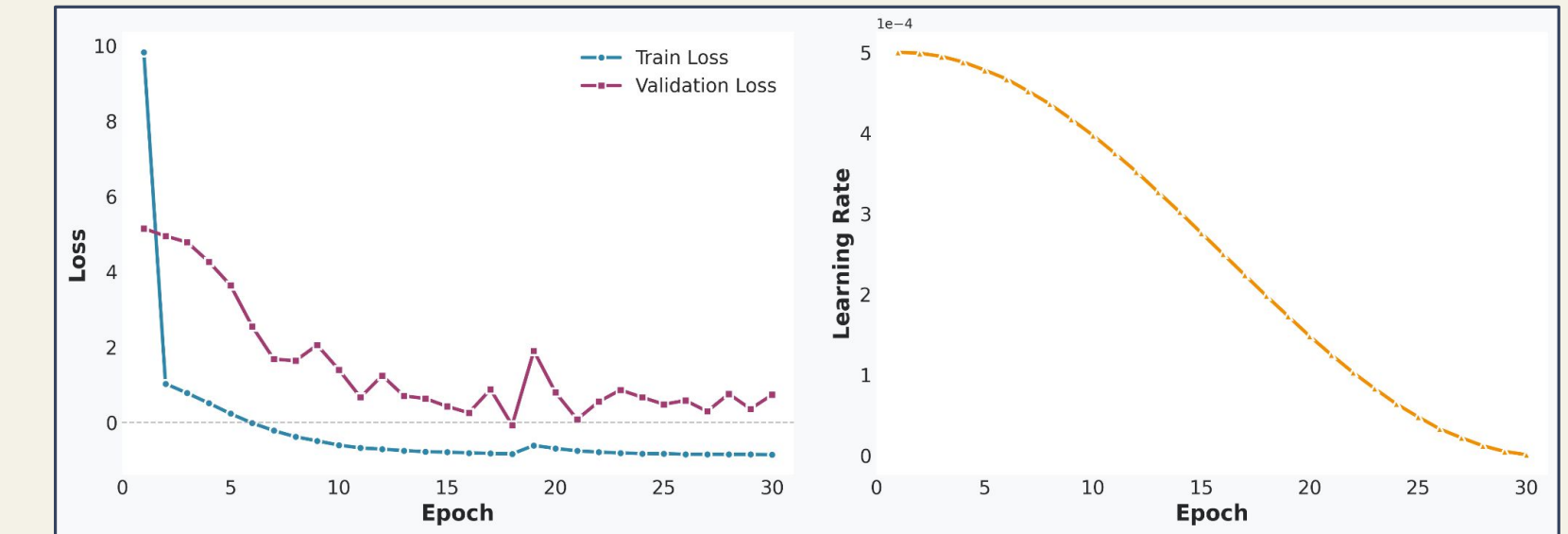
RMSE: Root-Mean Squared Error

- Model performance in capturing spatial or climate patterns

ACC: Anomaly Correlation Coefficient

- Model's ability to predict deviations from normal conditions

RESULTS



Pre-training ClimODE for 30 epochs on crops from Jan-Jun 2020

Variable	Metric	HRRR	HR-Extreme	HR-Extreme (Fine-tuned)
t2m	RMSE	17.01	14.85	11.52
	ACC	0.43	0.43	0.47
u10	RMSE	11.79	11.77	7.78
	ACC	0.22	0.35	0.50
v10	RMSE	8.66	8.88	5.93
	ACC	0.26	0.41	0.51
z	RMSE	130.98	159.36	121.30
	ACC	0.46	0.27	0.28
t	RMSE	12.66	14.36	9.98
	ACC	0.50	0.24	0.31

Results of fine-tuning on a subset of extreme events from July-Dec 2020

Variables

t2m: surface temp; u10, v10: 10-meter wind components ; z: geopotential; t: atmospheric temp

CONCLUSION

- Overall low-performance due to small training set
- Performance on extreme weather was worse than general weather for v10, z, and t variables
- Fine-tuned model improved performance on all metrics for extreme weather
→ can leverage ClimODE and fine tuning to improve extreme-weather performance

LIMITATIONS & FUTURE WORK

- Training set: Short-time period and cropped grids
→ use long-term data with larger grids
- Compare with purely data-driven GNN and transformer methods to evaluate if physics-informed model has an advantage in predicting extreme weather
- Explore equivariant geometric/graph models that better handle vector quantities and spatial structure.